CRYPTOGRAPHY & NETWORK SECURITY (CS3222-1)

Unit II – Task: Worked Solutions

Name: _____ **Roll No.:** _____

Notation. All arithmetic is done modulo the prime given in the problem; z^{-1} denotes the modular inverse of z. In the elliptic-curve problems Δ denotes the slope used in the point-addition and point-doubling formulas.

Problem 1: Diffie–Hellman Key Exchange and Man-in-the-Middle Attack (8 Marks)

Given: public parameters $p = 23$, $g = 5$; Alice's private key $a = 6$; Bob's private key $b = 15$.

(a) Public values exchanged by Alice and Bob *[2 marks]*

Alice: $A = g^a \bmod p = 5^6 \bmod 23$. Since $5^2 = 25 \equiv 2 \pmod{23}$,

$$A = (5^2)^3 \equiv 2^3 = 8 \pmod{23}.$$

Bob: $B = g^b \bmod p = 5^{15} \bmod 23$. By repeated squaring,

$$5^1 \equiv 5, \quad 5^2 \equiv 2, \quad 5^4 \equiv 2^2 = 4, \quad 5^8 \equiv 4^2 = 16 \pmod{23}.$$

As $15 = 8 + 4 + 2 + 1$,

$$B = 5^8 \cdot 5^4 \cdot 5^2 \cdot 5^1 \equiv 16 \cdot 4 \cdot 2 \cdot 5 = 640 = 27 \cdot 23 + 19 \equiv 19 \pmod{23}.$$

Alice sends $A = 8$ to Bob; Bob sends $B = 19$ to Alice.

(b) Shared secret key calculated at both ends *[3 marks]*

At Alice's end (Bob's public value $B = 19$, her private key $a = 6$):

$$K_A = B^a \bmod p = 19^6 \bmod 23.$$

$19^2 = 361 \equiv 16$ and $19^4 \equiv 16^2 = 256 \equiv 3$, hence

$$K_A = 19^4 \cdot 19^2 \equiv 3 \cdot 16 = 48 = 2 \cdot 23 + 2 \equiv 2 \pmod{23}.$$

At Bob's end (Alice's public value $A = 8$, his private key $b = 15$):

$$K_B = A^b \bmod p = 8^{15} \bmod 23.$$

$8^2 = 64 \equiv 18$, $8^4 \equiv 18^2 = 324 \equiv 2$ and $8^8 \equiv 2^2 = 4$, hence (with $15 = 8 + 4 + 2 + 1$)

$$K_B = 8^8 \cdot 8^4 \cdot 8^2 \cdot 8 \equiv 4 \cdot 2 \cdot 18 \cdot 8 = 1152 = 50 \cdot 23 + 2 \equiv 2 \pmod{23}.$$

Check: both equal $g^{ab} \bmod p = 5^{90} \bmod 23$; by Fermat, $5^{22} \equiv 1$, so $5^{90} = 5^{4 \cdot 22 + 2} \equiv 5^2 \equiv 2$.

$K_A = K_B = 2$, i.e. the shared secret key is **2**.

(c) Man-in-the-middle attack and one prevention mechanism *[3 marks]*

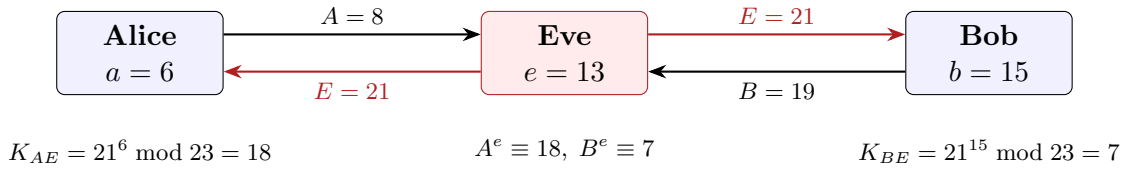
Plain Diffie–Hellman protects against a *passive* eavesdropper, but it provides **no authentication**: Alice cannot tell that the value she receives really came from Bob, and vice versa. An active attacker Eve who can intercept and replace the public values exploits this:

1. Eve chooses her own secret e and computes $E = g^e \bmod p$ (she may use a different secret for each victim; one is enough).
2. Alice sends $A = g^a \bmod p$. Eve intercepts it and sends E to Bob in its place.
3. Bob sends $B = g^b \bmod p$. Eve intercepts it and sends E to Alice in its place.
4. Alice computes $K_{AE} = E^a \bmod p = g^{ae} \bmod p$; Eve computes the same key as $A^e \bmod p$. Bob computes $K_{BE} = E^b \bmod p = g^{be} \bmod p$; Eve computes the same key as $B^e \bmod p$.
5. Alice believes K_{AE} is shared with Bob, and Bob believes K_{BE} is shared with Alice, but each actually shares a key with Eve. Eve decrypts every message with one key, reads or modifies it, re-encrypts it with the other key and forwards it, so neither user notices.

Numerical illustration (Eve chooses $e = 13$): $E = 5^{13} \bmod 23 = 5^8 \cdot 5^4 \cdot 5 \equiv 16 \cdot 4 \cdot 5 = 320 \equiv 21$. Using $21 \equiv -2 \pmod{23}$ and $2^{11} \equiv 1 \pmod{23}$:

$$K_{AE} = 21^6 \equiv (-2)^6 = 64 \equiv 18, \qquad K_{BE} = 21^{15} \equiv -2^{15} \equiv -2^4 = -16 \equiv 7 \pmod{23}.$$

Eve obtains the same two keys as $A^e = 8^{13} \equiv 18$ and $B^e = 19^{13} \equiv 7$. Since $18 \neq 7$, Alice and Bob do *not* share a key with each other, while Eve knows both.



Red arrows: values forged by Eve. Alice ends up with key 18, Bob with key 7 (both known to Eve).

Prevention (one effective mechanism): authenticated Diffie–Hellman using digital signatures and certificates. Each party signs its DH public value (together with a nonce or timestamp) using its private signing key and sends its X.509 certificate, issued by a trusted Certificate Authority. The receiver verifies the certificate and then the signature before accepting the value. Eve cannot forge a signature on E in Alice's or Bob's name, so the substitution is detected and the exchange is aborted. (This is the principle behind the station-to-station protocol and TLS.)

Problem 2: ElGamal Encryption and Decryption (8 Marks)

Given: prime $p = 23$, generator $g = 5$, receiver's private key $x = 6$, message $m = 10$, ephemeral (random) key $k = 3$.

(a) Receiver's public key *[2 marks]*

$$h = g^x \bmod p = 5^6 \bmod 23.$$

Since $5^2 \equiv 2 \pmod{23}$, we get $h = (5^2)^3 \equiv 2^3 = 8$.

Public key $(p, g, h) = (23, 5, 8)$; private key $x = 6$ (kept secret).

(b) Ciphertext pair (C_1, C_2) *[3 marks]*

$$C_1 = g^k \bmod p = 5^3 \bmod 23 = 125 = 5 \cdot 23 + 10 \equiv 10,$$
$$C_2 = m \cdot h^k \bmod p = 10 \cdot 8^3 \bmod 23.$$

Here $8^3 = 512 = 22 \cdot 23 + 6 \equiv 6$, therefore

$$C_2 = 10 \cdot 6 = 60 = 2 \cdot 23 + 14 \equiv 14 \pmod{23}.$$

Ciphertext $(C_1, C_2) = (10, 14)$.

(c) Decryption and verification *[3 marks]*

Step 1 – compute the shared mask:

$$s = C_1^x \bmod p = 10^6 \bmod 23.$$

$10^2 = 100 = 4 \cdot 23 + 8 \equiv 8$, so $10^6 = (10^2)^3 \equiv 8^3 = 512 \equiv 6$. Hence $s = 6$.

Step 2 – modular inverse of s: we need $6 \cdot s^{-1} \equiv 1 \pmod{23}$. Since $6 \cdot 4 = 24 \equiv 1$, we get $s^{-1} = 4$.

Step 3 – recover the message:

$$m = C_2 \cdot s^{-1} \bmod p = 14 \cdot 4 = 56 = 2 \cdot 23 + 10 \equiv 10 \pmod{23}.$$

Why it works: $s = C_1^x = (g^k)^x = (g^x)^k = h^k$, which is exactly the mask $h^k \equiv 6$ used during encryption, so $C_2 \cdot s^{-1} = m\, h^k\, h^{-k} = m$.

Decrypted message $m = 10$, which equals the original message ✓

Problem 3: Elliptic Curve Point Addition and Scalar Multiplication (8 Marks)

Given: $E: y^2 \equiv x^3 + 2x + 2 \pmod{17}$, i.e. $p = 17$, $a = 2$, $b = 2$; point $P = (5, 1)$.

Formulas used (point arithmetic on $E_p(a, b)$)

For $P = (x_1, y_1)$ and $Q = (x_2, y_2)$, the sum $R = P + Q = (x_3, y_3)$ is

$$x_3 = (\Delta^2 - x_1 - x_2) \bmod p, \qquad y_3 = \big(\Delta\,(x_1 - x_3) - y_1\big) \bmod p,$$

where the slope Δ is

$$\Delta = \begin{cases} \dfrac{y_2 - y_1}{x_2 - x_1} \bmod p, & P \neq Q \quad \text{(point addition)}, \\ \dfrac{3x_1^2 + a}{2y_1} \bmod p, & P = Q \quad \text{(point doubling; then } x_3 = \Delta^2 - 2x_1\text{)}. \end{cases}$$

Division means multiplication by the modular inverse.

(a) Verify that $P = (5, 1)$ lies on the curve *[2 marks]*

A point (x, y) lies on E if $y^2 \equiv x^3 + 2x + 2 \pmod{17}$. For $P = (5, 1)$:

$$\text{LHS} = y^2 = 1^2 = 1,$$
$$\text{RHS} = x^3 + 2x + 2 = 5^3 + 2 \cdot 5 + 2 = 125 + 10 + 2 = 137 = 8 \cdot 17 + 1 \equiv 1 \pmod{17}.$$

LHS = RHS, hence $P \in E$ ✓

(The curve is also non-singular: $4a^3 + 27b^2 = 4 \cdot 8 + 27 \cdot 4 = 140 \equiv 4 \not\equiv 0 \pmod{17}$*.)*

(b) Point doubling $2P = P + P$ *[3 marks]*

Here $x_1 = 5$, $y_1 = 1$, $a = 2$, $p = 17$. The doubling slope is

$$\Delta = \frac{3x_1^2 + a}{2y_1} = \frac{3 \cdot 5^2 + 2}{2 \cdot 1} = \frac{77}{2} \pmod{17}.$$

Now $77 = 4 \cdot 17 + 9 \equiv 9$ and $2^{-1} \equiv 9 \pmod{17}$ (because $2 \cdot 9 = 18 \equiv 1$). Hence

$$\Delta \equiv 9 \cdot 9 = 81 = 4 \cdot 17 + 13 \equiv 13 \pmod{17}.$$

Then

$$x_3 = \Delta^2 - 2x_1 = 13^2 - 2 \cdot 5 = 169 - 10 = 159 = 9 \cdot 17 + 6 \equiv 6 \pmod{17},$$
$$y_3 = \Delta\,(x_1 - x_3) - y_1 = 13\,(5 - 6) - 1 = -14 \equiv 3 \pmod{17}.$$

$2P = (6,\, 3)$. Check: $3^2 = 9$ and $6^3 + 2 \cdot 6 + 2 = 230 = 13 \cdot 17 + 9 \equiv 9$, so $2P \in E$.

(c) Point addition $3P = 2P + P$ *[3 marks]*

Take $(x_1, y_1) = 2P = (6, 3)$ and $(x_2, y_2) = P = (5, 1)$. The points are different ($x_1 \neq x_2$), so the addition slope is used:

$$\Delta = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 3}{5 - 6} = \frac{-2}{-1} = 2 \pmod{17}.$$

$$x_3 = \Delta^2 - x_1 - x_2 = 4 - 6 - 5 = -7 \equiv 10 \pmod{17},$$
$$y_3 = \Delta\,(x_1 - x_3) - y_1 = 2\,(6 - 10) - 3 = -11 \equiv 6 \pmod{17}.$$

$3P = (10,\, 6)$. Check: $6^2 = 36 \equiv 2$ and $10^3 + 2 \cdot 10 + 2 = 1022 = 60 \cdot 17 + 2 \equiv 2$, so $3P \in E$.

Problem 4: Elliptic Curve Diffie–Hellman (ECDH) Key Exchange (8 Marks)

Given: $E: y^2 \equiv x^3 + 2x + 2 \pmod{17}$, base point $P = (5, 1)$; Alice's private scalar $a = 3$; Bob's private scalar $b = 5$.
From Problem 3: $2P = (6, 3)$ and $3P = (10, 6)$.

(a) Public points of Alice and Bob *[3 marks]*

Alice: $Q_A = aP = 3P = (10, 6)$ (already obtained in Problem 3(c)).

Bob: $Q_B = bP = 5P$. Continue from $3P = (10, 6)$.

$4P = 3P + P = (10, 6) + (5, 1)$:

$$\Delta = \frac{1 - 6}{5 - 10} = \frac{-5}{-5} = 1,$$
$$x_3 = 1^2 - 10 - 5 = -14 \equiv 3, \qquad y_3 = 1 \cdot (10 - 3) - 6 = 1 \quad \Rightarrow \quad 4P = (3, 1).$$

$5P = 4P + P = (3, 1) + (5, 1)$:

$$\Delta = \frac{1 - 1}{5 - 3} = 0,$$
$$x_3 = 0^2 - 3 - 5 = -8 \equiv 9, \qquad y_3 = 0 \cdot (3 - 9) - 1 = -1 \equiv 16 \quad \Rightarrow \quad 5P = (9, 16).$$

Cross-check: $2P + 3P = (6,3) + (10,6)$ gives $\Delta = \frac{3}{4} \equiv 3 \cdot 13 \equiv 5$, $x_3 = 25 - 6 - 10 = 9$, $y_3 = 5(6-9) - 3 = -18 \equiv 16$, i.e. $(9,16)$ again.

n	1	2	3	4	5
nP	$(5,1)$	$(6,3)$	$(10,6)$	$(3,1)$	$(9,16)$

Alice's public point $Q_A = 3P = (10, 6)$; Bob's public point $Q_B = 5P = (9, 16)$.

(b) Shared point computed independently with both private scalars *[3 marks]*

Alice computes $S_A = a\,Q_B = 3\,(9,16)$; Bob computes $S_B = b\,Q_A = 5\,(10,6)$.

Alice: $S_A = 3Q_B = 2Q_B + Q_B$ with $Q_B = (9,16)$.

Doubling $2Q_B$:

$$\Delta = \frac{3 \cdot 9^2 + 2}{2 \cdot 16} = \frac{245}{32} \equiv 7 \cdot 15^{-1} \equiv 7 \cdot 8 = 56 \equiv 5 \pmod{17} \qquad (15 \cdot 8 = 120 \equiv 1),$$
$$x_3 = 5^2 - 2 \cdot 9 = 7, \qquad y_3 = 5\,(9-7) - 16 = -6 \equiv 11 \quad \Rightarrow \quad 2Q_B = (7,11).$$

Addition $2Q_B + Q_B = (7,11) + (9,16)$:

$$\Delta = \frac{16 - 11}{9 - 7} = \frac{5}{2} \equiv 5 \cdot 9 = 45 \equiv 11 \pmod{17},$$
$$x_3 = 11^2 - 7 - 9 = 105 \equiv 3, \qquad y_3 = 11\,(7-3) - 11 = 33 \equiv 16 \quad \Rightarrow \quad S_A = (3,16).$$

Bob: $S_B = 5Q_A = 4Q_A + Q_A$ with $Q_A = (10,6)$.

Doubling $2Q_A$:

$$\Delta = \frac{3 \cdot 10^2 + 2}{2 \cdot 6} = \frac{302}{12} \equiv 13 \cdot 12^{-1} \equiv 13 \cdot 10 = 130 \equiv 11 \pmod{17} \qquad (12 \cdot 10 = 120 \equiv 1),$$
$$x_3 = 11^2 - 2 \cdot 10 = 101 \equiv 16, \qquad y_3 = 11\,(10-16) - 6 = -72 \equiv 13 \quad \Rightarrow \quad 2Q_A = (16,13).$$

Doubling $4Q_A = 2(2Q_A)$:

$$\Delta = \frac{3 \cdot 16^2 + 2}{2 \cdot 13} = \frac{770}{26} \equiv 5 \cdot 9^{-1} \equiv 5 \cdot 2 = 10 \pmod{17} \qquad (9 \cdot 2 = 18 \equiv 1),$$
$$x_3 = 10^2 - 2 \cdot 16 = 68 \equiv 0, \qquad y_3 = 10\,(16-0) - 13 = 147 \equiv 11 \quad \Rightarrow \quad 4Q_A = (0,11).$$

Addition $4Q_A + Q_A = (0,11) + (10,6)$:

$$\Delta = \frac{6 - 11}{10 - 0} = \frac{-5}{10} \equiv 12 \cdot 10^{-1} \equiv 12 \cdot 12 = 144 \equiv 8 \pmod{17} \qquad (10 \cdot 12 = 120 \equiv 1),$$
$$x_3 = 8^2 - 0 - 10 = 54 \equiv 3, \qquad y_3 = 8\,(0-3) - 11 = -35 \equiv 16 \quad \Rightarrow \quad S_B = (3,16).$$

Cross-check: P has order 19 (one finds $9P = (7,6)$, $10P = (7,11)$, hence $19P = \mathcal{O}$), so $15P = -4P = (3,-1) = (3,16)$.

$S_A = S_B = 15P = (3, 16)$. The shared point is $(3,16)$; its x-coordinate, **3**, is typically used as the shared secret.

(c) Why the shared point matches and the scalars stay secret *[2 marks]*

Same shared point. Scalar multiplication on the group of curve points is associative and the scalars commute ($ab = ba$), so both parties reach $(ab)P$ (in the example, $15P = (3,16)$):

$$S_A = a\,Q_B = a(bP) = (ab)P = (ba)P = b(aP) = b\,Q_A = S_B.$$

Secrecy of a and b. Only P, $Q_A = aP$ and $Q_B = bP$ travel over the channel; the scalars a and b are never transmitted. To recover a from P and aP (or b from P and bP) an attacker must solve the *elliptic-curve discrete logarithm problem* (ECDLP); computing abP from aP and bP alone is the *elliptic-curve Diffie–Hellman problem*. For a curve with a large prime-order subgroup (e.g. 256-bit) no efficient algorithm is known, so the scalars and the shared point remain hidden. (The toy curve here has only 19 points and could be brute-forced instantly; it only illustrates the mechanism.)

Summary of final answers

1 $A = 8$, $B = 19$, shared key $K = 2$; MITM prevented by authenticated DH (digital signatures / certificates).

2 $h = 8$ (public key $(23, 5, 8)$), $(C_1, C_2) = (10, 14)$, decrypted $m = 10$ ✓

3 $P \in E$ ✓, $2P = (6, 3)$, $3P = (10, 6)$.

4 $Q_A = 3P = (10, 6)$, $Q_B = 5P = (9, 16)$, shared point $15P = (3, 16)$.